# Load Safety Gate

One photo at the gate. GO or NO-GO before the truck leaves.

Mode: Startup · Track: Computer Vision · Team Destroyerrrrs

GO WARN NO-GO

Body roll · Overhang · Height · Plates and people blurred before anything is shown

# A badly loaded truck is checked by eye, or not at all

## Who uses it

Gate operator or dispatcher at a terminal, fleet depot or dealer receiving yard. One photo, one answer, before the truck leaves.

## What goes wrong today

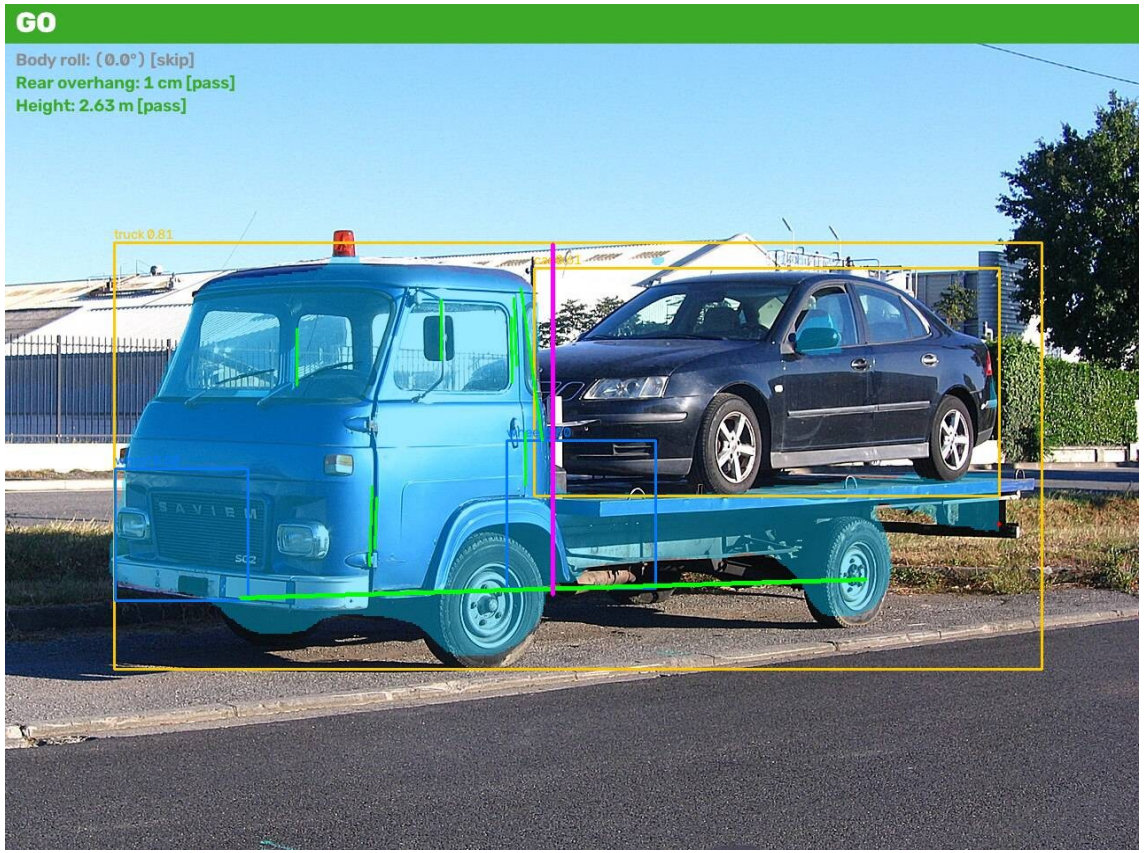
One-sided loads roll over. Overhanging cargo hits bridges and other vehicles. Oversized loads draw fines and stop the truck at the first police post.

## Who pays

Fleet owner or terminal, per gate. Every Toyota and BYD arrives on a car carrier into a Royal Motors yard, and car carriers are the classic over-height and rear-overhang offenders.

## Regulation already exists

Uzbekistan Decree 342, Annex 2: height 4.0 m, width 2.55 m, rear protrusion 2 m. Nobody measures it at the gate.




Live output of the deployed pipeline on a licensed test image (Wikimedia Commons, CC BY-SA 4.0).

# Live demo

## amir-kuldashev--load-safety-gate-web.modal.run

Open it on your phone. Take a photo of any truck, or give us one. No login, no app.

- Plates and people are pixelated first. The original is never stored.
- Green lines: body vertical edges and the ground line the roll is measured against.
- Red line: cargo beyond the body edge, in centimetres. Magenta: cargo top to ground, in metres.
- Verdict banner, per-check confidence, and the measured latency on CPU.
- Every run gets an inspection ticket and lands in the shift journal: counts of GO, WARN, NO-GO for the day. Numbers only, the photo is discarded.
- When the photo does not allow a measurement the check says so. No guess, ever.



# A pipeline, not a prompt

1

## Photo

phone or gate camera

2

## Detect

YOLO-World boxes: truck, trailer, cargo, car, plate, person. YOLOv8-seg truck mask. Pretrained, no training.

3

## Blur

plates and people pixelated before storage or display

4

## Measure

wheel finder, ground line, body verticals vs scene verticals, scale from 2.55 m body or 1.05 m wheel

5

## Decide

Decree 342 thresholds; GO, WARN, NO-GO, or CANNOT ASSESS

6

## Report

annotated image, checklist, confidence, latency

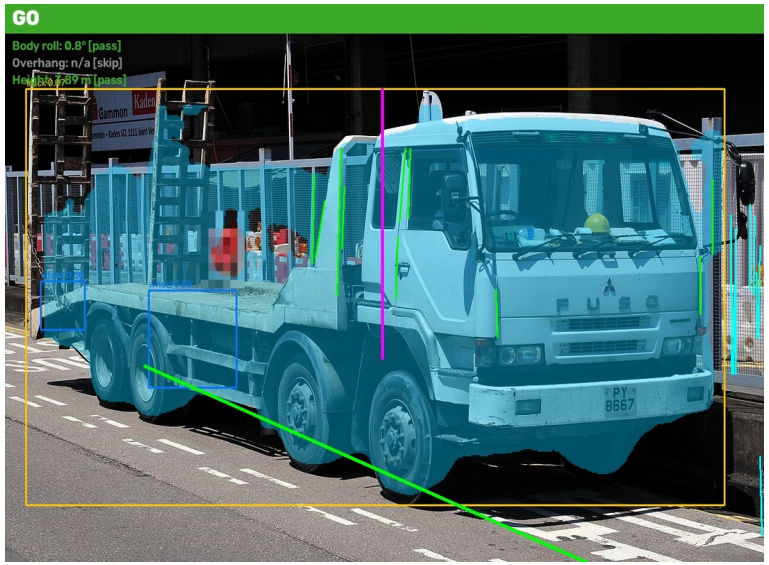
### Ours

Privacy stage, wheel finder, geometry, decision rules, the 301-image licensed set, the 120-image labelled evaluation, the app and the deployment.

### Borrowed, with attribution

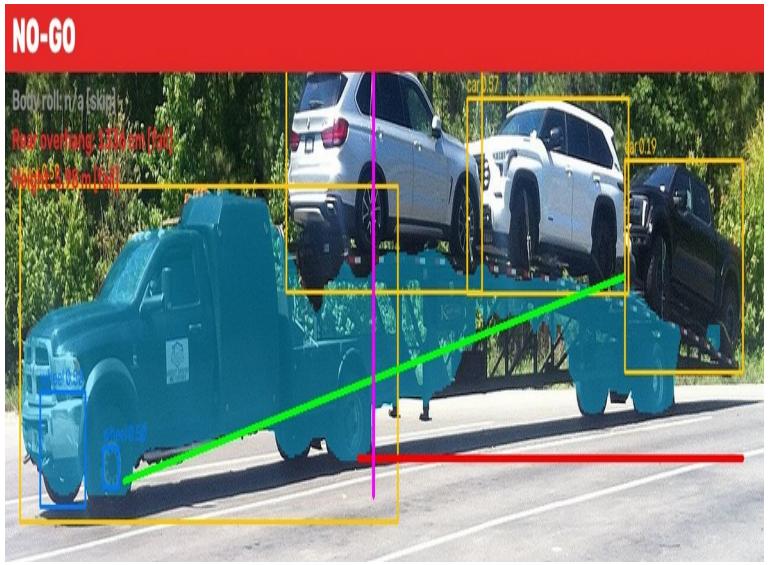
Two pretrained Ultralytics models (AGPL-3.0). Open-vocabulary prompts, no fine-tuning needed for the demo. Runs on CPU in 0.19 s per image.

# Measured relative to the truck itself, so a phone photo is honest




## Body roll

Green: the body's vertical edges. Cyan: vertical lines in the scene. The angle between them is the roll, so a tilted phone does not tilt the truck.




## Overhang

Red: cargo beyond the body edge, converted to centimetres with the 1.05 m wheel or the 2.55 m body width in the same photo.




## Height

Magenta: cargo top to the ground line through the tyres, in metres, against the 4.0 m limit.

Every number is relative to an object in the same photo. No camera calibration, no absolute pose. When the photo has no usable reference the check says CANNOT ASSESS.

Photos: Wikimedia Commons, Ralf Roletschek (CC BY 3.0); Bull-Doser (Public domain); Bob Adams from George, South Africa (CC BY-SA 2.0). Annotations drawn by the pipeline.

# Measured on a test set we built, including the bad news

| Subset | Images | Truck found | Wheels | Scale | Roll reported |
|---|---|---|---|---|---|
| Car carriers | 70 | 0.89 | 0.60 | 0.71 | 0.21 |
| Flatbeds | 55 | 0.87 | 0.71 | 0.76 | 0.27 |
| Semi-trailers | 30 | 0.97 | 0.70 | 0.83 | 0.37 |
| Oversized loads | 44 | 0.86 | 0.50 | 0.59 | 0.11 |
| Night | 38 | 0.97 | 0.66 | 0.74 | 0.45 |

Detector hit rate on 301 licensed Commons photos (no labels needed). Night holds up: the truck mask is the most robust stage.

| Check | Precision | Recall | tp/fp/fn |
|---|---|---|---|
| Tilted body | 0.40 | 0.18 | 2/3/9 |
| Rear overhang | 0.00 | 0.00 | 0/1/7 |
| Side overhang | n/a | 0.00 | 0/0/6 |
| Over height | 0.57 | 0.24 | 4/3/13 |
| Clean trucks, no false alarm | 39 of 44 | | 89% specificity |

Against 101 team-labelled images, 61 of them three-quarter views. Honest result: the phone mode is conservative. 89% of clean trucks pass without a false alarm, but it caught only 6 of 41 labelled positives, mostly three-quarter views where no scale object is visible. The calibrated gate camera is the fix, not a bigger model.

**1.63°**

roll instrument error, 56 rotated level photos at known angles, sign right 93% of the time

**0.19 s**

per image on 2 CPU cores, measured on the deployed server

**< $0.001**

compute per inspection; one warm server about $90 per month

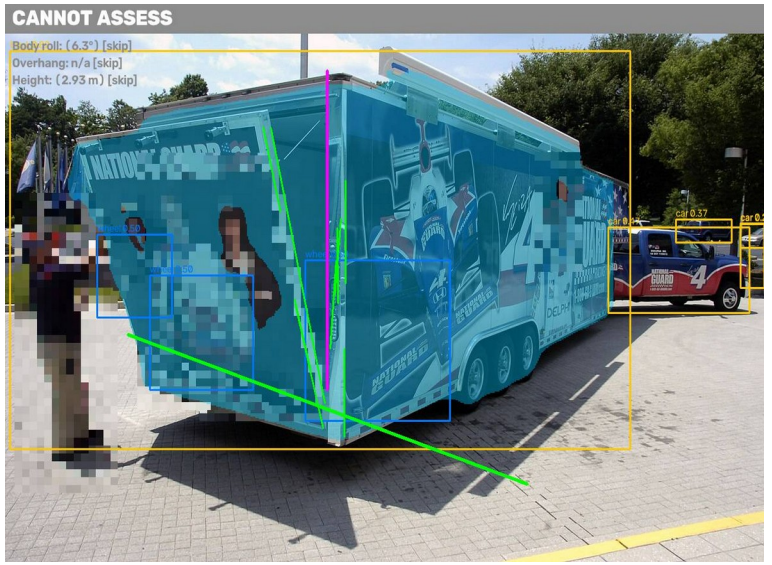
**0 faces**

stored. Plates and people pixelated before anything is kept

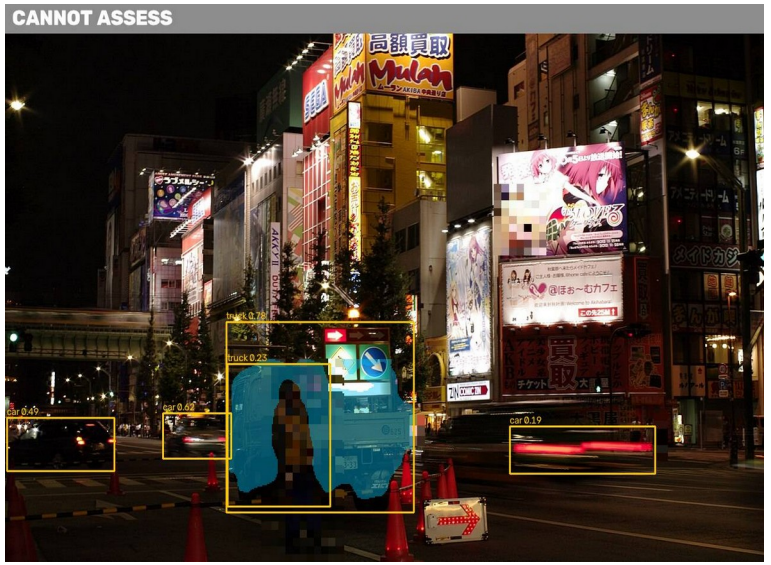
# Where it breaks, and what it does instead of guessing




Not a truck, or a tiny truck in the frame. Result: CANNOT ASSESS. The overloaded-vehicles category on Commons is mostly carts and motorbikes.




Front or three-quarter views foreshorten the body. Roll confidence drops below the floor and the check is skipped.




Night street with no truck in frame: CANNOT ASSESS. Wheels hidden in shadow or behind tanks give a wrong scale; a height that comes out above 8 m is shown as unreliable, never reported as a measurement.

## Design rule

Every check reports a confidence and is skipped below a floor. A skipped check is shown as such, never as a pass.

# The replication test

## A weekend with the same tools gets you

- A detector that draws boxes around trucks and cargo
- A prompt to a vision model that says "looks overloaded"
- No measurement, no threshold, no number a gate operator can act on

## What they do not get

- Perspective-safe roll from vertical edges, with a gravity reference in the same image
- Centimetres and metres from a known object in the frame, and a CANNOT ASSESS path
- A decision layer written from Decree 342, the regulation a police post actually applies
- A privacy stage that makes the product legal to pilot
- 301 licensed images with hit rates, 120 labelled, and a deployed gate product

# Pilot budget and payback, every assumption on the slide

| Pilot cost, 3 months | Amount | Assumption |
|---|---|---|
| Team: 3 people, 3 months | $7,200 | $800 per person per month |
| Cameras and mounts, 3 gates | $600 | $200 per gate, one-off |
| Hosting | $270 | one warm CPU server, $90 per month |
| Labelling and data | $500 | about 40 hours of labelling |
| Legal and privacy review | in team time | blur at ingest, no personal data stored, one-day legal check |

| Avoided loss, one fleet | Amount | Assumption, calibrated in the pilot |
|---|---|---|
| Departures screened | 200 per day | mid-size fleet, one gate camera |
| Dangerous departures caught | 1 per month | conservative: 1 in ~6,000 departures |
| Cost of one load incident | $5,000 | cargo loss + downtime + fine + claim handling; wide range |
| Avoided loss per year | $60,000 | 12 incidents x $5,000 |

## Unit economics

Price per gate per month, software only. Compute per inspection is below $0.001, so margin is in the subscription, not the inference. One server serves dozens of gates.

## Payback

Pilot cost $8,570 (team $7,200 + cameras $600 + hosting $270 + labelling $500). Avoided loss $5,000 per month at one caught incident a month. Payback = 8,570 / 5,000 = 1.7 months, inside the pilot itself. The fleet's own incident rate replaces the assumption after month one.

# Next steps

## Month 1

Fixed, calibrated camera at one gate. Reference marker removes scale ambiguity. Ten-second video mode.

## Month 2

Fine-tune the detector on our labelled set. Add open door, loose strap and missing marking classes.

## Month 3

Weighbridge and dispatch integration. Pilot at a Royal Motors receiving yard on incoming car carriers.

Repository: github.com/Winner-121/Destroyerrrrs_Final_Stage · Demo: amir-kuldashev--load-safety-gate-web.modal.run

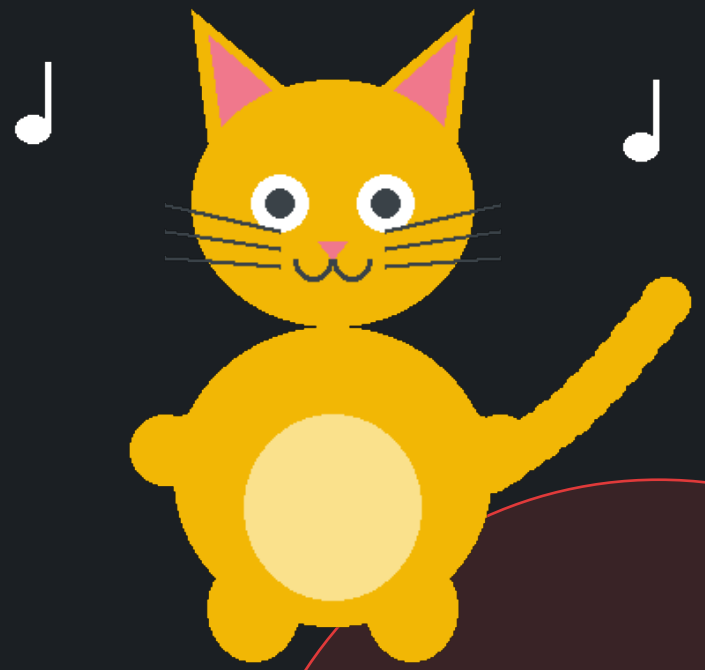
# Questions?

Open the gate on your phone right now:

`amir-kuldashev--load-safety-gate-web.modal.run`

`github.com/Winner-121/Destroyerrrrs_Final_Stage`

**GO** **WARN** **NO-GO**



Team Destroyerrrrs, thank you